

Shock

- **Acute clinical syndrome:** A sudden and severe health condition requiring immediate attention.
- **Systemic reduction in tissue perfusion (hypoperfusion):** Decreased blood flow to tissues throughout the body.
- **Tissue injury and organ dysfunction:** Damage to tissues and failure of organ functions.

Pathophysiology of Shock

- **Decrease in effective circulating volume:** Reduction in the amount of blood or fluid circulating through the body.
- **Decreased cardiac output:** Reduced amount of blood pumped by the heart per minute.
- **Hypoxia/Anoxia:** Low or no oxygen in tissues.

Pathogenesis

1. Cellular Ischemia:

- **Hypoperfusion:** Decreased blood flow to tissues.
- **ATP production:** Energy produced by cells.
- **Lysosomal enzyme damage:** Breakdown of enzymes inside cells leading to irreversible damage.
- **Lactic acid:** Byproduct of anaerobic metabolism, leading to metabolic acidosis.
- **Flowchart:**
 - **Na/K pump:** A pump in cells maintaining the balance of sodium and potassium ions.
 - **Anaerobic metabolism:** Energy production without oxygen.
 - **Capillary damage:** Injury to small blood vessels.

2. Inflammatory Mediators:

- **Septic shock:** Shock caused by infection and inflammation.
- **Endotoxin:** Toxins released by bacteria into the bloodstream.
- **Cytokines:** Proteins that regulate the immune response.
- **TNF- α , IL-1 β :** Specific cytokines involved in inflammation.
- **Prostaglandins:** Lipid compounds that cause inflammation and other reactions.
- **Complement fragments:** Particles that are part of the immune system response (C3a, C5a).

3. Free Radical Injury:

- **Adenosine, inosine, hypoxanthine:** Molecules that build up during ischemia.
- **Reperfusion:** The restoration of blood flow after ischemia.
- **Superoxide (O₂⁻), hydrogen peroxide (H₂O₂):** Harmful molecules formed during reperfusion.

- **Hydroxyl radicals:** Highly reactive molecules that can cause cellular damage.
- **Neutrophils:** White blood cells involved in immune response, which can cause further tissue damage.

1. Non-progressive (Compensated, Reversible) Shock

- **Hypoxia:** Insufficient oxygen supply to tissues.
- **Mitochondria:** Organelles in cells that produce energy (ATP).
- **Anaerobic respiration:** Energy production in the absence of oxygen.
- **Lactic acid and pyruvic acid:** Byproducts of anaerobic metabolism.
- **Metabolic acidosis:** A condition where the body becomes too acidic due to excess acid production.
- **Redistribution of blood:** Reallocation of blood flow to essential organs (brain, heart).

2. Progressive (Decompensated) Shock

- **Adrenergic discharge:** Release of adrenaline-related hormones to prepare the body for fight or flight.
- **Vasoconstriction:** Narrowing of blood vessels to increase blood pressure.
- **Splanchnic viscera:** Internal organs in the abdominal cavity, such as the stomach and intestines.
- **Hyperventilation:** Rapid and deep breathing to remove excess carbon dioxide and increase oxygen intake.
- **Metabolic acidosis:** See above.
- **Venous return:** The flow of blood back to the heart.
- **Left ventricle:** The chamber of the heart responsible for pumping oxygenated blood to the body.
- **Stroke volume:** The amount of blood pumped by the heart with each beat.
- **Vasoconstrictive hormones:** Hormones that cause blood vessels to constrict.
- **Renin-Angiotensin-Aldosterone System (RAAS):** A hormone system that regulates blood pressure and fluid balance.
- **ADH (Antidiuretic hormone):** A hormone that helps the body retain water.
- **Epinephrine:** A hormone (adrenaline) that prepares the body for quick action.
- **Resorption of fluid:** Movement of fluids back into the bloodstream from tissues.
- **Interstitial space:** The space between cells and tissues.
- **Intravascular pressure:** Pressure within the blood vessels.
- **Intracellular to extracellular fluid shift:** Movement of fluid from inside the cells to the surrounding tissues.
- **Renal water conservation:** The process by which the kidneys retain water to prevent dehydration.

3. Irreversible (Decompensated) Shock

- **Vasoconstriction:** See above.
- **Vascular permeability:** The ability of blood vessels to allow substances to pass through them.
- **Severe hypoxia:** Extreme lack of oxygen in tissues.
- **Cell death:** When cells stop functioning and die due to lack of oxygen and nutrients.
- **MODS (Multiple Organ Dysfunction Syndrome):** A life-threatening condition where multiple organs fail simultaneously due to severe illness or injury.

1. Hypovolemic Shock

- **Circulatory volume:** The total amount of blood circulating in the body.
- **Venous return:** The flow of blood back to the heart.
- **Cardiac output:** The amount of blood the heart pumps per minute.
- **SVR (Systemic Vascular Resistance):** The resistance of the blood vessels to blood flow.

2. Cardiogenic Shock

- **Dysfunction of heart ventricles:** The inability of the heart's lower chambers to pump blood effectively.
- **Myocardial infarction (MI):** Heart attack caused by blockage of blood flow to heart muscles.
- **Cardiac arrhythmias:** Abnormal heart rhythms.
- **Pulmonary embolism:** Blockage in a pulmonary artery, usually caused by a blood clot.

3. Distributive Shock

- **Vasodilation:** The widening of blood vessels.
- **Maldistribution of blood flow:** Abnormal distribution of blood to tissues or organs.
- **Endotoxins:** Toxins released by bacteria, often causing widespread inflammation.
- **Gram-negative bacteria:** A group of bacteria that do not retain the crystal violet stain used in Gram staining, often more harmful.
- **Sympathetic tone:** The activity of the sympathetic nervous system that regulates involuntary functions like blood vessel constriction.
- **Spinal injury:** Damage to the spinal cord that can disrupt nerve function.
- **Spinal anesthesia:** A type of anesthesia where medication is injected into the spinal fluid, often causing loss of sensation and muscle control.
- **Hypersensitivity (Type I):** An exaggerated immune response, commonly seen in allergies.

4. Obstructive Shock

- **Physical obstruction:** A blockage in the blood vessels or heart.

- **Cardiac tamponade:** A condition where fluid builds up in the sac surrounding the heart, restricting its ability to pump blood.

Signs and Symptoms of Shock

1. CNS Symptoms

- **Cerebral perfusion:** Blood flow to the brain.
- **Hypoxia:** Insufficient oxygen supply to tissues.

2. Cardiovascular Symptoms

- **Hypotension:** Low blood pressure.
- **Tachycardia:** Abnormally fast heart rate.
- **Weak, thready pulse:** A pulse that is weak and difficult to feel, often due to low blood flow.

3. Skin Changes

- **Vasoconstriction:** Narrowing of blood vessels, often in response to cold or stress.
- **Mottled extremities:** Skin appears patchy, often due to poor blood flow.
- **Pallor:** Unusual paleness of the skin, often due to decreased blood flow.

4. Respiratory Symptoms

- **Tachypnea:** Abnormally rapid breathing.
- **Acidosis:** A condition where the body becomes too acidic, often due to a buildup of carbon dioxide or lactic acid.

5. Other Symptoms

- **Hypothermia:** A dangerously low body temperature.
- **Sweat evaporation:** The process of sweating, which cools the body down.
- **Fatigue:** Extreme tiredness, often due to inadequate oxygen supply.
- **Fainting:** Temporary loss of consciousness due to insufficient blood flow to the brain.
- **Oliguria/anuria:** Reduced urine output (oliguria) or complete lack of urine (anuria), often due to poor kidney perfusion.
- **Afferent arteriolar vasoconstriction:** Narrowing of the small blood vessels leading to the kidneys, reducing blood flow to them.

A. Resuscitation

- **Airway ventilation:** Ensuring the air passages are open for adequate air supply.
- **Oxygenation:** The process of adding oxygen to the blood.

- **Endotracheal intubation:** Inserting a tube into the trachea to ensure proper airflow during emergencies.

B. Control of Hemorrhage

- **Hypovolemic shock:** Shock caused by a significant loss of blood or fluids.
- **Elevation:** Raising a body part to reduce bleeding.
- **Compression bandage:** A tight bandage used to stop bleeding.
- **Ligation:** Tying off a blood vessel to prevent bleeding.
- **Internal bleeding:** Bleeding that occurs inside the body, often not visible externally.
- **Resuscitation:** Restoring the body's circulation and oxygen levels to a safe state.

C. Extracellular Fluid Replacement

- **Fluid replacement:** Administering fluids to restore the body's normal volume.
- **CVP (Central Venous Pressure):** Measurement of pressure in the veins returning blood to the heart, used to monitor fluid levels.
- **Normal saline:** A sterile solution of salt in water used for fluid replacement.
- **Ringer's lactate:** A type of fluid that helps restore electrolytes and hydration.
- **Blood substitutes:** Fluids used when blood is unavailable, like plasma or dextran.

D. Correct Acid-Base Disturbance

- **Sodium bicarbonate:** A compound used to neutralize excess acid in the blood.
- **Electrolyte abnormalities:** Imbalances in essential minerals (like potassium or calcium) in the body.

E. Drugs

1. Sedatives

- **Pain and anxiety relief:** Reducing discomfort and worry during medical procedures.
- **Morphine:** A potent pain-relieving drug.
- **Pethidine:** Another type of pain relief drug.
- **IV (Intravenous):** Administration of drugs directly into a vein.
- **IM (Intramuscular):** Injection into a muscle.

2. Chronotropic Agents

- **Low heart rate:** A slower than normal heart rate.
- **Atropine:** A medication that increases heart rate.
- **Isoproterenol:** A drug that can increase heart rate and cardiac output.

3. Inotropic Agents

- **Cardiac muscle contraction:** The contraction of heart muscles to pump blood.
- **Dopamine:** A drug that stimulates the heart and improves blood circulation.
- **Dobutamine:** A drug that strengthens heart muscle contraction.
- **Myocardial activity:** The functioning and contraction of the heart muscle.
- **Renal blood flow:** Blood flow to the kidneys.
- **Vasodilation:** The widening of blood vessels to increase blood flow.

4. Vasodilators

- **Septic shock:** Shock caused by infection and inflammation.
- **Traumatic shock:** Shock resulting from physical injury or trauma.
- **Cardiogenic shock:** Shock caused by the heart's inability to pump blood.
- **Nitroprusside:** A drug that helps relax blood vessels and improve blood flow.
- **Nitroglycerin:** A drug that helps widen blood vessels and reduce heart workload.

5. Vasoconstrictors

- **Neurogenic shock:** Shock caused by damage to the nervous system.
- **BP (Blood pressure):** The pressure exerted by circulating blood on the walls of blood vessels.
- **Myocardial activity:** The function and contractions of the heart muscle.
- **Phenylephrine:** A drug that narrows blood vessels to increase blood pressure.
- **Metaraminol:** A drug that helps increase blood pressure by constricting blood vessels.

6. Beta-Blockers

- **Stiff myocardium:** Reduced flexibility or function of the heart muscle.
- **Cardiogenic shock:** See above.
- **Propranolol:** A medication that helps control heart rate and reduce the heart's workload.

7. Diuretics

- **Cardiogenic shock:** See above.
- **Vascular volume:** The amount of blood circulating in the blood vessels.
- **Filling pressure:** The pressure exerted on the heart chambers when they fill with blood.
- **Hypovolemic shock:** See above.
- **Septic shock:** See above.

Definition of Hemorrhage

- **Rupture:** Breaking or bursting of a blood vessel.

Types of Hemorrhage

I. Based on the Type of Blood Vessel

- **Arterial:** Related to arteries, which carry oxygenated blood.
- **Venous:** Related to veins, which carry deoxygenated blood.
- **Capillary:** Tiny blood vessels connecting arteries and veins.
- **External:** Occurs outside the body.
- **Internal:** Happens within the body, not visible externally.

II. Based on Timing

- **Reactionary:** Occurring as a delayed response to an initial injury.
- **Ligature slip:** When a tied-off blood vessel becomes untied.
- **Vasospasm:** Sudden narrowing of blood vessels.
- **Secondary:** Happens later, often due to infection.
- **Sloughing:** Shedding or peeling away of tissue.

III. Based on Blood Loss (ATLS Classification)

- **Class I-IV:** Categories based on the percentage of total blood volume lost.

Management of Hemorrhage

1. General Management (Hospitalization)

- **Isotonic saline:** A solution with the same salt concentration as body fluids.
- **Balanced crystalloid:** Fluids containing electrolytes to maintain balance.
- **Sedation:** Use of medications to calm the patient.

2. Specific Measures to Stop Hemorrhage

A. Conservative Methods

- **Tourniquet:** A device used to compress blood vessels and stop blood flow.
- **Elevation:** Raising a body part to reduce blood flow.

B. Operative Methods

- **Clamping:** Closing a blood vessel with a clip or tool.
- **Thermocautery/Diathermy:** Use of heat or electrical current to seal blood vessels.
- **Adrenaline Swabs:** Swabs soaked in adrenaline to constrict blood vessels.
- **Silver Clips:** Metal clips used in surgeries to control bleeding.
- **Suture:** Stitching a wound or blood vessel.
- **Muscle Graft:** Using muscle tissue to cover or stop bleeding.

- **Bone Wax:** A material used to stop bleeding from bones.
- **BIPP Paste:** Paste with antiseptic and wax to control bone bleeding.
- **Dacron Mesh:** A synthetic material used in vascular repair.
- **Organ Excision:** Surgical removal of an organ (e.g., splenectomy for spleen).

3. Restoration of Blood Volume

- **Ringer's lactate:** A fluid containing electrolytes used to replace lost fluids.
- **Hyperchloremic acidosis:** Increased blood acidity due to high chloride levels.
- **Hypotonic dextrose solutions:** Fluids with lower salt concentration, avoided in hemorrhage.
- **Packed Red Blood Cells (PRBCs):** Concentrated red cells used for transfusions.
- **Thrombocytopenia:** Low platelet count in the blood.

Septic Shock

Pathophysiology

- **Endotoxins:** Toxic substances released by bacteria, causing harmful effects.
- **Lipopolysaccharides (LPS):** Molecules in the outer membrane of Gram-negative bacteria.
- **Monocytes/Macrophages:** Types of white blood cells involved in immune defense.
- **Toll-like receptor (TLR-4):** A protein that recognizes pathogens and activates immune responses.
- **Cytokines:** Proteins that regulate inflammation and immunity (e.g., IL-1, TNF).
- **Endothelial injury:** Damage to the inner lining of blood vessels.
- **Leukocyte adhesion:** White blood cells sticking to blood vessel walls.
- **DIC (Disseminated Intravascular Coagulation):** A condition causing widespread blood clotting and bleeding.
- **Myocardial failure:** Inability of the heart to pump blood effectively.

Clinical Features

- **Pyrexia:** Medical term for fever.
- **Rigors:** Shivering caused by fever or infection.
- **Hypotension:** Abnormally low blood pressure.
- **Cerebral perfusion:** Blood flow to the brain.
- **Hyperventilation:** Rapid or deep breathing.
- **Respiratory alkalosis:** Imbalance in blood pH due to excessive carbon dioxide exhalation.
- **Medullary stimulation:** Activation of the brainstem region controlling respiration.

Treatment of Septic Shock

Infection Control

- **Antibiotics:** Medications to kill or inhibit the growth of bacteria.
- **Surgical debridement:** Removal of infected or dead tissue.
- **Radiological drainage:** Using imaging to guide the removal of pus or fluid from the body.

Fluid Replacement

- **Crystalloids:** Solutions of minerals and water used for hydration (e.g., normal saline, lactated Ringer's solution).
- **Colloids:** Solutions containing larger molecules (e.g., albumin, dextrans) to maintain blood volume.
- **Pulmonary capillary wedge pressure (PCWP):** Measurement indicating fluid pressure in the heart.

Vasopressor Therapy

- **Vasopressors:** Drugs that constrict blood vessels to raise blood pressure (e.g., dopamine, norepinephrine).
- **Hemodynamic stability:** Maintaining consistent blood flow and pressure in the body.

Vasoactive Drugs

- **Phenoxybenzamine:** A medication that relaxes blood vessels.
- **Isopropranolol:** A drug affecting heart rate and contraction strength.

Mechanical Ventilation

- **Endotracheal intubation:** Inserting a tube into the airway to assist breathing.
- **Ventilation:** Using a machine to provide oxygen to the lungs.

Steroid Therapy

- **Hypoadrenalism:** A condition where adrenal glands don't produce enough hormones.
- **Methylprednisolone/Dexamethasone:** Steroid drugs to reduce inflammation and support adrenal function.

Pathophysiology

- **Cardiac output (CO):** The amount of blood the heart pumps in one minute.
- **Baroreceptors:** Nerve endings that sense changes in blood pressure.
- **Peripheral vascular resistance (PVR):** Resistance to blood flow in small blood vessels.
- **Tachycardia:** Abnormally fast heart rate.
- **Arteriolar constriction:** Narrowing of small arteries.
- **Epinephrine & norepinephrine:** Hormones that increase heart rate and constrict blood vessels.
- **ACTH (Adrenocorticotropic Hormone):** A hormone that stimulates stress response.
- **ADH (Antidiuretic Hormone):** Hormone that helps retain water in the body.
- **Renin-angiotensin system:** A hormone system regulating blood pressure and fluid balance.
- **Angiotensin II:** A molecule causing blood vessel constriction.
- **Aldosterone:** Hormone that promotes sodium and water retention.
- **Anaerobic metabolism:** Energy production without oxygen.
- **Acidosis:** Excess acid in the body fluids.
- **Mediator release:** Chemicals causing inflammation.

Clinical Features

- **Hypotension:** Low blood pressure.
- **Tachypnea:** Rapid and shallow breathing.
- **Hypothermia:** Dangerously low body temperature.
- **Oliguria:** Decreased urine output.
- **Anuria:** Absence of urine output.
- **Mottled skin:** Irregular skin discoloration due to poor circulation.

Classification of Hemorrhagic Shock

- **Pulse:** Heartbeats per minute.
- **Respiratory rate:** Number of breaths per minute.
- **Urine output:** Volume of urine excreted.
- **Mental state:** Level of awareness or responsiveness.

Management of Primary Hemorrhage

- **Clots:** Semi-solid masses of coagulated blood.
- **Socket:** The hollow space left after tooth extraction.
- **Gauze:** A thin, woven fabric used for dressing wounds.

If Bleeding Continues

- **Anaesthetize:** To numb an area to block pain.
- **Nerve block:** An injection near a nerve to block sensation in a larger area.
- **Local infiltration:** Injection of anesthesia directly into the area of treatment.
- **Lidocaine:** A common local anesthetic.
- **Epinephrine:** A medication added to anesthetics to constrict blood vessels and reduce bleeding.
- **Curette:** A surgical instrument for scraping tissue or bone.
- **Irrigate:** To wash out with a fluid.
- **Suture:** Stitching a wound to close it.
- **Haemostatic agents:** Substances that promote clotting and stop bleeding.
 - **Oxidized cellulose:** A material used to control bleeding.
 - **Topical thrombin:** A clot-promoting enzyme applied directly to bleeding tissue.
 - **Microfibrillar collagen:** A protein-based agent aiding in blood clot formation.

Neurogenic Shock

- **Distributive shock:** A type of shock where blood vessels lose their ability to constrict, leading to poor circulation.
- **Sympathetic tone:** Normal activity of the sympathetic nervous system maintaining blood vessel constriction.
- **Sympathetic nervous system:** Part of the autonomic nervous system that increases heart rate and constricts blood vessels.

Causes of Neurogenic Shock

- **Spinal cord neoplasm:** Tumor affecting the spinal cord.
- **Epidural anesthesia:** A regional anesthesia injected into the epidural space around the spinal cord.

Pathophysiology

- **Baroreceptor response:** Reflex controlling blood pressure by adjusting heart rate and vessel tone.
- **Autoregulation:** The ability of tissues to regulate their own blood supply.
- **Vasospasm:** Sudden constriction of blood vessels.
- **Thrombosis:** Formation of a blood clot inside a vessel.
- **Neurotransmitter:** Chemical messenger in the nervous system.
- **Free radicals:** Unstable molecules causing damage to cells.

Clinical Features

- **Bradycardia:** Abnormally slow heart rate.
- **Peripheral vasoconstriction:** Narrowing of blood vessels in the extremities.
- **Tachypnea:** Rapid breathing.

Management of Neurogenic Shock

- **Airway patency:** Keeping the airway open and clear.
- **Intravascular volume:** Blood volume within blood vessels.
- **Vasoconstrictors:** Medications that narrow blood vessels to increase blood pressure.
- **Capacitance:** The ability of blood vessels to store blood.
- **Venous return:** Blood flow back to the heart from the body.

Definition of Blood Transfusion

- **Transferring:** Moving from one person or place to another.

Types of Blood Transfusions

- **Homologous (Allogeneic):** Blood from another person.
- **Autologous:** Blood donated and stored by the same individual for later use.

Indications of Blood Transfusion

- **Acute Hemorrhage:** Sudden and severe bleeding.
- **Significant blood loss:** A substantial decrease in blood volume.
- **Severe Anemia:** A condition where hemoglobin levels are very low.
- **Hemoglobin:** A protein in red blood cells that carries oxygen.
- **Blood Disorders:** Conditions affecting the components or function of blood.
- **Hemophilia:** A disorder causing poor blood clotting.
- **Leukemia:** Cancer of blood-forming tissues, such as bone marrow.
- **Thalassemia:** A genetic disorder causing abnormal hemoglobin production.
- **Erythroblastosis Fetalis:** A severe form of anemia in newborns caused by Rh incompatibility.
- **Rh incompatibility:** A mismatch between mother and fetus blood types in the Rh factor.
- **Exchange transfusion:** Replacing the infant's blood with donor blood.
- **Chemotherapy:** Treatment of cancer using drugs.
- **Suppress:** To reduce or stop an activity.
- **Uremia:** High levels of waste products in the blood due to kidney failure.
- **Hypertension:** High blood pressure.

Resuscitation

- **Airway ventilation:** Process of ensuring the airway is open and air reaches the lungs.
- **Oxygenation:** Supplying oxygen to the blood and tissues.
- **Intubation:** Inserting a tube into the airway to maintain breathing.

Fluid Resuscitation

- **Large gauge needle/cannula:** A wide needle or tube used to administer fluids quickly.
- **Monitor CVP:** Checking **Central Venous Pressure**, which measures blood returning to the heart.

Fluids Used

- **Lactated Ringer's Solution:** A solution of electrolytes that replenishes body fluids.
- **Hypertonic Saline:** A saltwater solution with higher salt concentration than body fluids.
- **Hetastarch:** A synthetic substance that increases blood volume.
- **Colloid:** A fluid containing particles, used for plasma expansion.
- **Dextran:** A carbohydrate used as a plasma expander to restore blood volume.
- **Blood substitutes:** Products like plasma or dextran used temporarily in place of blood.
- **DIC (Disseminated Intravascular Coagulation):** A condition causing widespread blood clotting and bleeding, often triggered by mismatched blood transfusions.

Shock and Triage in Trauma

1. **Shock:** A medical emergency where the body is unable to supply enough blood to vital organs, leading to insufficient oxygenation and potential organ failure.
2. **Trauma:** Physical injury or damage caused by an external force such as accidents, falls, or violence.
3. **Prioritize:** To arrange or deal with patients based on the urgency and severity of their conditions.
4. **Mass Casualty Incidents:** Situations where there are a large number of patients, often due to disasters, accidents, or large-scale emergencies, requiring urgent medical attention.
5. **Vital Organs:** Organs essential for life, such as the **heart, brain, lungs, and kidneys**.
6. **Resource Allocation:** The process of distributing medical resources, such as staff, equipment, and medications, based on patient needs in emergencies.
7. **Hypotension:** Low blood pressure, which is a common feature in shock.

8. **Airway Management:** Techniques used to ensure that a patient's airway remains clear and unobstructed, especially in trauma cases.
9. **Avulsion:** The tearing away of a part of the body, such as the skin, from its normal attachment.
10. **Fractures:** Broken bones, which can lead to shock if severe and not treated promptly.
11. **START (Simple Triage and Rapid Treatment):** A method used in emergency care to quickly assess and categorize patients based on the severity of their conditions.
12. **SALT (Sort, Assess, Life-saving interventions, Treatment/Transport):** A method for triaging patients during mass casualty incidents, which includes sorting patients, assessing their needs, providing life-saving interventions, and determining treatment and transport.
13. **MIMMS (Major Incident Medical Management and Support):** A structured approach used for triaging patients during large-scale incidents and ensuring proper medical management and support.

Vasovagal Shock

Causes of Vasovagal Shock

- **Vagal nerve:** A nerve that helps control heart rate and blood pressure.
- **Syncope:** Fainting or temporary loss of consciousness due to a drop in blood flow to the brain.
- **Overstimulation:** Excessive activation or excitement of a nerve or system.

Treatment of Vasovagal Shock

- **Trendelenburg position:** A body position where the patient is lying flat with their legs elevated above the head to improve circulation.
- **Intravenous (IV):** A method of delivering fluids or medications directly into the bloodstream through a vein.
- **Hypoxic:** A condition where there is not enough oxygen in the body or tissues.

Management of Vasovagal Shock

- **Vital signs:** Basic body measurements like heart rate, blood pressure, and temperature used to monitor health.
- **Recurrence:** The reappearance of a condition after it was previously treated or resolved.
- **Posture:** The way in which someone holds their body, especially when standing or sitting.

Key Points

- **Condition:** A state of health or a particular illness or situation.
- **Reassure:** To make someone feel less worried or anxious, especially about their health.